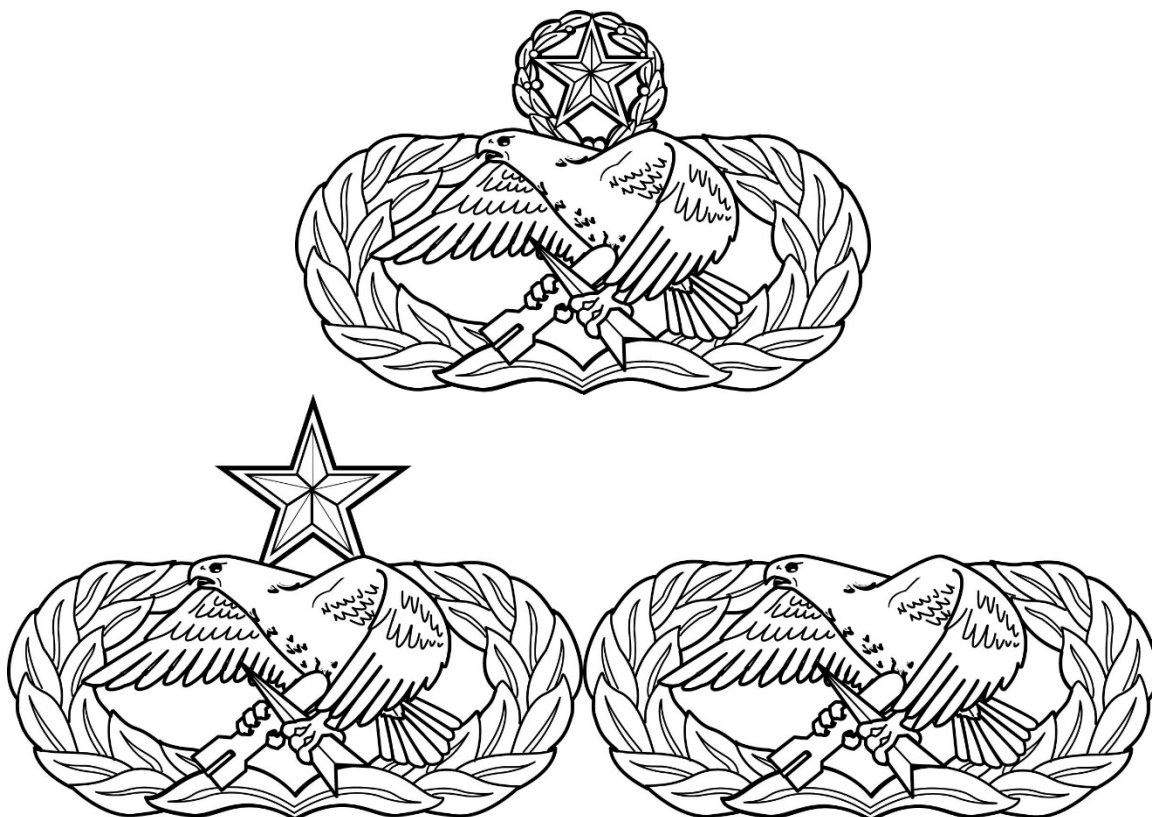


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PRECISION MEASUREMENT EQUIPMENT LABORATORY



CAREER FIELD EDUCATION AND TRAINING PLAN
(CFETP)

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CAREER FIELD EDUCATION AND TRAINING PLAN PRECISION MEASUREMENT EQUIPMENT LABORATORY AFSC 2P0X1

PART I

PREFACE

1. Preface. The CFETP is a comprehensive education and training document that identifies life cycle education and training requirements, training support resources, and minimum core task requirements for this specialty. The CFETP provides personnel a clear career path to success and instills rigor in all aspects of career field training. Note: Civilians occupying associated positions use Part 2 to support duty position qualification training.
2. The CFETP consists of two parts; supervisors plan, manage, and control training within the specialty using both parts of the plan.
 - 2.1. **Part 1** provides information necessary for overall management of the specialty. Section A explains how individuals will use the plan; section B identifies career field progression information, duties and responsibilities, training strategies, and career field path; section C associates each level with specialty qualifications (knowledge, education, experience, training, and other); and section D indicates resource constraints. Some examples are funds, manpower, equipment, facilities; section E identifies transition training guide requirements for E-5 through E-7.
 - 2.2. **Part 2** includes the following: Section A: identifies the STS and includes duties, tasks, Technical References to support training, AETC conducted training, wartime course and core task and correspondence course requirements. Section B: contains the course objective list and training standards supervisors use to determine if service members satisfied training requirements. Section C: identifies available support materials. An example is a qualification training package, which may be developed to support proficiency training; section D identifies a training course index supervisors can use to determine resources available to support training. Included here are both mandatory and optional courses; section E identifies MAJCOM unique training requirements supervisors can use to determine additional training required for the associated qualification needs. At unit level, supervisors and trainers use Part 2 to identify, plan, and conduct training commensurate with the overall goals of this plan.
3. Using CFETP guidance ensures individuals in this specialty receive effective and efficient training at the appropriate points in their career. This plan enables us to train today's work force for tomorrow's jobs.

SECTION A – GENERAL INFORMATION***PURPOSE***

4. This CFETP provides the information necessary for CFMs, MFMs, commanders, UTM, supervisors and trainers to plan, develop, manage, and conduct an effective and efficient career field training program. The plan outlines the training individuals in this Air Force specialty should receive to develop and progress throughout their career. This plan identifies initial skills, upgrade, qualification, advanced and proficiency training. Initial skills training is the Air Force specialty specific training an individual receives upon entry into the Air Force or upon retraining into this specialty for award of the 3-skill level. Normally, this training is conducted by AETC at one of the technical training centers. Upgrade training identifies the mandatory courses, task qualification requirements, and correspondence course completion requirements for award of the 3-, 5-, 7-, 9-skill level. Qualification training is actual hands-on task performance training designed to qualify a member in a specific duty position. This training program occurs both during and after the upgrade training process. It is designed to provide the performance skills and knowledge required to do the job. Advanced training is formal specialty training used for selected service members. Proficiency training is additional training, either in-residence or exportable advanced training courses, or OJT provided to personnel to increase their skills and knowledge beyond the minimum required for upgrade. The CFETP has several purposes—some are:

- 4.1. Serves as a management tool to plan, manage, conduct, and evaluate a career field's training program. Also, it is used to help supervisors identify training at the appropriate point in an individual's career.
- 4.2. Identifies task and knowledge training requirements for each skill-level in the specialty and recommends education and training throughout each phase of an individual's career.
- 4.3. Lists training courses available in the specialty, identifies sources of training, and the training delivery method.
- 4.4. Identifies major resource constraints that impact full implementation of the desired career field training process.

USE OF THE CFETP

5. The plan is used by MFMs and supervisors at all levels to ensure comprehensive and cohesive training programs are available for everyone in the specialty.

- 5.1. AETC training personnel develop or revise formal resident, nonresident, field, and exportable training based upon requirements established by the users and documented in Part 2 of the CFETP. They also work with the CFM to develop acquisition strategies for obtaining resources needed to provide the identified training.
- 5.2. MFMs ensure their training programs complement the CFETP mandatory initial, upgrade, and proficiency requirements. OJT Training, resident training, and contract training

or exportable courses can satisfy identified requirements. Ensure MAJCOM-developed training to support this AFSC is identified for inclusion into the plan.

5.3. Each individual completes the mandatory training requirements specified in this plan. The list of courses in Part 2 is used as a reference to support training.

COORDINATION AND APPROVAL

6. The CFM is the approval authority. Also, the CFM will initiate an annual review of this document to ensure currency and accuracy. (T-1) MAJCOM representatives and AETC training personnel will identify and coordinate on the career field training requirements. (T-2) Using the list of courses in Part 2, ensures elimination of duplicate training.

SECTION B – CAREER PROGRESSION AND INFORMATION

SPECIALTY DESCRIPTION

7. Performs and manages repair, calibration, and modification of test, measurement, and diagnostic equipment (TMDE), including precision measurement equipment laboratory (PMEL) standards and automatic test equipment. Supervises the process and use of TMDE to perform electronic and physical measurement competencies in support of local, regional, and enterprise-wide sustainment.

DUTIES AND RESPONSIBILITIES

8. For more information refer to the Department of the Air Force Enlisted Classification Directory (DAFECD)

8.1. Inspects, aligns, troubleshoots, and repairs PMEL standards, common and weapon system peculiar TMDE. Inspects TMDE for preventive maintenance, cleanliness, and safety requirements. Performs equipment maintenance using theories of operation, block diagrams, schematics, logic trees, and software diagnostics. Isolates malfunctions to component level. Calibrates and certifies TMDE to technical data specifications traceable to the International System of Units. Records and reports maintenance data; ensures compliance with the AFMETCAL cyber security program, prepares technical order improvement reports, special training requests, training quality reports, and modification proposals. Tracks equipment warranties. Provides training and manages technical order distributions. Handles, labels, and disposes of hazardous materials and waste according to environmental standards.

Plans, organizes, and coordinates mission support and/or readiness requirements. Collects and analyzes maintenance data to mitigate risk with the PMEL in accordance with the AFMETCAL Risk Management program. Identifies mission essential TMDE and its impact on workload. Coordinates lateral support, AFMETCAL Repair Network Management Program, inter/intra-service agreements, and/or contract services. Evaluates procedures for storage, inventory, and inspection of property. Provides training and assistance to TMDE users.

8.2. Develops and evaluates workload plans, budget, and support agreements. Implements the PMEL Management System (MS) and Quality Program (QP). Manages PMEL resource

stewardship (facility, equipment, inventory, environmental control systems, etc.). Submits reports to higher headquarters and maintains a safe working environment. Maintains PMEL automated management systems (PAMS) system of record.

SKILL AND CAREER PROGRESSION

9. Adequate training and timely progression from the apprentice to the superintendent level play an important role in the Air Force's ability to accomplish its mission. It is essential that everyone involved in training do their part to plan, manage, and conduct an effective training program. The guidance provided in this part of the career field education and training plan ensures everyone receives viable training at appropriate points in their career.

9.1. Apprentice (3) Level. Upon completion of initial skills training Airmen will be awarded their 3-level. Trainees will work with trainers to enhance their knowledge and skills.

9.2. Journeyman (5) Level. Once upgraded to the 5-level, Journeymen enter continuation training to broaden their experience base. Journeymen may be assigned as trainers. Journeymen will complete supplemental courses, exportable courses, and MAJCOM specific training as required. They will utilize the Intermediate PMEL Metrology CDC, OJT, and Task Qualification Training to progress in their career field. Individuals should also consider continuing education by working towards a Community College of the Air Force (CCAF) degree and AF COOL program certificates. Wear of the Basic Maintenance Badge is authorized on award of the 5 skill-level.

9.3. Craftsman (7) Level. Once upgraded to 7-level, Craftsmen can expect to fill various supervisory and management positions. For specific information regarding responsibilities of these duty positions refer to DAFMAN 21-113, *Air Force Metrology and Calibration Program*. Craftsman may also be assigned to work in staff positions. Craftsman will complete supplemental courses, exportable courses, and MAJCOM specific training as required. They will utilize the Advanced PMEL Metrology CDC, OJT and Task Qualification Training to progress in their career field. Continued academic education through CCAF, higher degree programs, and AF COOL program certificates are encouraged. Wear of the Senior Maintenance Badge (star) is authorized on award of the 7-skill level.

9.4. Superintendent (9) Level. Once upgraded to 9-level, a Superintendent can be expected to fill the highest levels of positions within the Flight, MAJCOM or HAF. For specific information regarding responsibilities of these duty positions refer to DAFMAN 21-113. Additional training in the areas of budget, manpower, resources, and personnel management should be pursued through continuing education. Additional higher education and completion of courses outside their career AFSC are also recommended. They will utilize the Expert PMEL Metrology CDC (See Note, Para 18.7), OJT and Task Qualification Training to progress in their career field. Continued academic education through higher degree programs and AF COOL program certificates are encouraged. Wear of the Master Maintenance Badge (wreath and star) is authorized on award of the 9-skill level.

9.5. 2P0X1 Air Force Job Qualification Standard (AFJQS). If approved by the AFCFM, Airmen are expected to complete training outlined in the 2P0X1 AFJQS as they progress through their career. Supervisors at all levels determine what specific tasks are applicable for their assigned Airmen based on local mission, duty position and grade as appropriate. The 2P0X1 AFJQS offers a framework of discrete, tangible tasks that measure an Airmen's proficiency to perform specific tasks warranting specific task qualification. All personnel will be qualified to the GO/NO GO standards for all tasks listed in the 2P0X1 AFJQS. GO means the individual can perform all parts of the task utilizing appropriate instructions or other procedural guidance and needs only a spot check of completed work. Training listed in Part II of this instruction is provided through OJT, functional distance learning, and/or formal in-residence training courses. Supervisors/trainers will utilize the career field functional training products (e.g. Competency Assessment Workbooks, Training Plans, exportable training) when available, to conduct the training.

TRAINING DECISIONS

10. The career field education and training plan uses a building block approach (simple to complex) to encompass the entire spectrum of training requirements for the PMEL (2P0X1) career field. The spectrum includes a strategy for when, where, and how to meet the training requirements. The strategy should be apparent and affordable to reduce duplication of training and eliminate a disjointed approach to training. The following training decisions were made at the 2P0X1 Specialty Training Requirements Team (STRT) Meeting held at the Air Force Metrology & Calibration Program office at Heath, OH 6-10 May 2024.

10.1. The STRT was tasked with shifting the paradigm of how the 2P career field views training. The STRT was the final phase of the transition the career field training model from an equipment based CFETP and STS to a Competency Based training model. In 2018, the AFCFM completed a STRT to pursue a redesign of the CFETP and training focus to transition from an equipment-based training approach to a training model focused on developing metrology skills. However, due to the COVID-19/SARS2 pandemic and a lack of model to work on the transition was not completed. In 2022 the AFCFM coordinated with AETC/A3J, Competency Modeling Division to officially launch a complete overhaul of the training model.

10.2. On 1 March 2024 the AFCFM submitted a request supported by HAF/A4LM to 2 AF/A3T to begin the analysis to relocate the technical training pipeline from Keesler AFB, MS to Sheppard AFB, TX. The reasoning behind the decision is aligned with the CSAF Great Power Competition (GPC) Re-Optimization decision. Currently the technical training pipeline is assigned to the 335 TRS and 2P0X1 is the last Maintenance career field remaining on Keesler and has caused historical issues obtaining sustainment funding and modernization. Part of the GPC Re-Optimization decision is to redesignate AETC to Airmen Development Command and one part of the decision is to consider consolidation of Air Force Specialty training at co-located locations to ensure proper alignment within the respective Deputy Chief of Staff functional authority priorities and investment strategy, which for 2P0X1 is the HAF/A4. The relocation to Sheppard further supports decisions to develop multi-capable 2P0X1 Airmen able to augment 2W0X1 functions during a contingency or conflict.

10.3. In 2023 the USMC Ground Systems Command collaborated with the 2P0X1 AFCFM to pursue increased joint service interoperability. The AFCFM approved a USMC Ground Systems Marine to attend the Air Force PMEL Apprentice Course at Keesler AFB in June 2023. The success and feedback from the test drove the drafting and support for establishing a Interservice Requirements Organization (ITRO) agreement between the USAF and USMC to consolidate apprentice training at the PMEL Apprentice course at Keesler. The analysis for the ITRO is set to kick off May 2024 and will analyze not only the feasibility of the effort but the impacts if the USAF PMEL technical training relocates to Sheppard. Both USMC and USAF leadership supports the ITRO but a move to Sheppard may need to re-shape the ITRO to share costs between the services to ensure the USMC can continue to attend the USAF PMEL Apprentice course.

10.4. The CFETP has undergone a considerable revision towards building a competency-based training and development platform for the PMEL career field. A significant change has been to shift the focus from task-based training to one that is more centered on behavior/outcomes-based learning. A task is a unit of work activity or operation which forms a significant part of a duty. These are singular in nature and are usually accomplished in one continuous action, which also can occur independently of other tasks. Conversely, outcomes are learning goals that typically consist of a multitude of tasks. These outcomes are actions and performances that embody and reflect the learner's competence in using content, information, ideas, and tools successfully. Focusing on learning outcomes allow organizations, leaders, supervisors, and trainers to incorporate foundational competencies and underlying characteristics (values, traits, attitudes) into learning, which is necessary for developing Airmen with the competencies needed for future challenges. The following decisions were made with close coordination between HQ AETC, 2AF Technical Training, schoolhouse instructors and staff, field SMEs, functional managers and the AFCFM. The final training requirements are then approved by the AFCFM.

10.5. During the STRT held 6-10 May 2024. Members of the STRT finalized the competency model. This was accomplished by reverse engineering the behaviors found in the PMEL occupational competency model and then by asking what Airmen need to know and to master a specific behavior. The intent of the learning outcomes is to identify all factors needed to succeed in attaining the behavior. Formal training will be codified by using the behavioral statement coding system for the STS as opposed to the proficiency code key. As a result, each line item will consist of a verb and the coding system for formal training will only use P (performance), K (knowledge), and pk (performance-knowledge).

10.6. **Competencies.** The Air Force defines competencies as a combination of knowledge, skills, abilities, and other characteristics that manifest in observable and measurable pattern of behaviors required for mission success. The PMEL community has rebuilt the entire training platform by using competencies to identify the behaviors that are needed to be successful on the job. This has been accomplished by identifying and integrating the 2P, PMEL, occupational competencies and leveraging the Air Force's foundational competencies in a manner that provides all Airmen with transparent and unbiased pathways towards their own successful development.

Airmen's Foundational Competencies. The foundational competencies are a set of accepted and valued competencies, which enable success across a wide array of DAF missions, roles, functions, and duties. These competencies are the core of Airmen development and enable Airmen with tools, pathways, and capabilities to improve their performance in any job, specialty, or situation. The foundational competencies are grouped into different categories of Developing Self, Developing Others, Developing Ideas, and Developing Organization. Airmen can go to MyVector (accessible via AF Portal) to complete a self-assessment, which will have them evaluate themselves on the 24 Airmen's foundational competencies or a 360-degree assessment, where subordinates, peers, and leaders can also provide feedback. The assessment tools will provide Airmen with immediate feedback on personal strengths and areas for improvement. Additionally, a personal



improvement plan with targeted resources (videos, reading content, developmental opportunities) are provided for continued self-development.

Figure 1. Airmen's Foundational Competencies.

10.7. Occupational Competencies. Occupational competencies are a set of competencies required of all Airmen within a specific workforce category. These competencies provide a framework that describes the knowledge, skills, abilities, and other characteristics needed to perform that function's mission successfully.

10.7.1. A career field's competencies can be viewed in a competency model, which is an organized collection of competencies pertinent to the career field. The occupational competency model provides a framework to effectively assess, maintain, and monitor the competencies required for mission success for Airmen within the PMEL community. The occupational competency modeling process follows a distinct process with continued involvement from the career field and allows Airmen to see how their task lists, OJT, formal courses, and other training, education, and experiences are aligned to the career field's strategic objectives.

Figure 2. 2P0X1, The PMEL Occupational Competency Model

Competency	Sub-Competency	Sub-Competency Description
Maintenance Activities	Electrical Measurement	The methods, devices, and calculations utilized to measure electrical quantities and qualities.
	Physical Measurement	The methods, devices, and calculations utilized to measure physical/dimensional quantities and qualities.
	Alignment/Repair	The process of members performing equipment maintenance using theories of operation, block diagrams, schematics, logic trees, and software diagnostics to sustain TMDE.
Management System	Training	The process of providing a structural foundation of skills, methods, and characteristics to develop a qualified
	Logistics	Manage and coordinate all logistical activities related to the movement, storage, and maintenance of TDME, including inventory management, transportation, asset tracking, and customer relations.
	Supervision	Individuals or teams lead to ensure efficient and effective operations and adherence to regulations that achieve organizational goals.
	Quality Program	Members systematically assess to sample PMEL product quality, procedural compliance, and effectiveness.

10.7.2. Career fields work with trained competency experts to identify and develop their competency model, which consists of the competencies, sub-competencies, and definitions. Occupational competency models will be different for each career field. The model focuses on integrating not just the technical components, but also leadership, management, combat, joint, all-domain, and social mastery competencies required for Airmen to succeed in their career field.

10.7.3. After a model is developed, a team of subject matter experts begin building competency rubrics, which consists of the competency, a description of the competency, proficiency levels, and measurable and observable behaviors. The competency rubrics will help Airmen learn which behaviors are aligned to the career field's strategic direction, the professional developmental expectations, and the criteria for success. Figure 3 provides an example of a competency rubric for 2P career field.

Figure 3. 2P0X1, The PMEL Occupational Competency model for Electrical Measurement

Competency	Proficiency Levels	Observable Behaviors
Test, Measurement, and Diagnostic Equipment Maintenance (Certification)	Expert Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency	- Develops new processes across Measurement Area Disciplines within the element to enhance the measurement capability - Collaborates with external agencies engineers to validate and develop alternate calibration methods
Sub-Competency	Advanced	- Identifies deficiencies and takes corrective actions across multiple Measurement Area Disciplines within the Electronics Element - Determines deviations from procedural steps when substitute equipment is required - Maintains measurement standards traceability to Air Force Primary Standards Laboratory
Electrical Measurement	Consistency of Application: Sustained application of competency over time in complex situations	
Description	Intermediate	- Maximizes measurements across multiple Measurement Area Disciplines within the Electronics Element to calibrate Test, Measurement and Diagnostic Equipment - Calculates tolerances of measurement results to determine the Test, Measurement and Diagnostic Equipment condition
The methods, devices, and calculations utilized to measure electrical quantities and qualities.	Consistency of Application: Sustained application of competency over time in a variety of situations	
Supporting Competencies	Basic	- Performs measurements utilizing metrology principles to calibrate Test, Measurement and Diagnostic Equipment - Utilizes measurement standards to apply established tolerances to maintain established test uncertainty ratios and determine Test, Measurement and Diagnostic Equipment condition - Applies general maintenance practices to produce safe, accurate, reliable, and traceable Test, Measurement and Diagnostic Equipment
Analytical Thinking Critical Thinking Precision Information Seeking	Consistency of Application: Sustained application of competency over time	

10.7.4. To better understand how to read and utilize the competency rubric, a breakdown of each component is explained below in Figure 4.

Figure 4. Competency Rubric Example

Competency		The competency section states the competency group.
Management System		
Sub-Competency		The sub-competency section states the narrower category that forms part of the competency group. Note: Some models may only consist of a competency and not include a sub-competency.
Quality Program		
Description		The description section provides a statement that gives details about the sub-competency, enabling career field members to better understand how sub-competency relates to the AFS.
Members systematically assess to sample PMEL product quality, procedural compliance, and effectiveness.		
Supporting Competencies		The supporting competencies section are supported-level competencies that are linked to the success of the sub-competency. These competencies lend themselves more toward areas like values, traits, and attitudes. These competencies were included as part of a larger survey that went out to the entire AFS; respondents were asked to rate the top supporting competencies they believe will attribute to higher successful performance within the sub-competency.
Analytical Thinking Communication Information Seeking Accountability		

10.7.5. To better understand how to read and utilize the competency rubric, a breakdown of each component is explained below in Figure 5.

Figure 5. Proficiency Levels Example

Proficiency Levels		The proficiency levels are broken into four parts: basic, intermediate, advanced, and expert.
Expert Impact on Wing /Group Level		Under each proficiency level are predetermined criteria selected by a group of SMEs from your career field and validated by the career field. The criteria were used as the basis to develop the observable behaviors. These criteria provide concrete parameters for the behaviors, which are consistent but progressive in nature as a member moves up the scale from basic to expert.
Advanced Impact on management decisions		Some of the criteria (e.g. depth of knowledge, consistency of application/complexity, and thinking challenge) allows an individual to become an expert through the experience gained in a particular job and over a period of time. For example, the person can quickly move up different proficiency levels while they are serving as a technician at a flight; they move quickly because they are exposed to a variety of situations.
Intermediate Impact on specific workplace projects		
Basic Impact on specific workplace tasks		While other criteria (e.g. scope, impact, and reach of influence) requires more of a hierarchical approach to gain the experience needed to progress through the competency levels. Moving through the proficiency levels may be difficult to do in certain jobs. For example, if scope at the expert level requires job integration with the AF-level, then the individual may have to be in a position where they can gain that experience (i.e. at HHQ, Wing, or an organization with far reaching capabilities).

Figure 6. Observable Behaviors Example

Observable Behaviors		The observable behaviors are statements of what can be observed from an individual manifesting the competency at the respective competency level. They provide objective evidence that the individual possesses the competency level, and shows what effective performance looks like. The behaviors are written to be specific enough so they can be observable and lend themselves towards measurement.
- Conducts Air Force Metrology and Calibration assessments to determine the effectiveness and the capability to perform Safe, Accurate, Reliable and Traceable measurements to the National Institute of Standards and Technology or other approved sources - Implements the Air Force Metrology and Calibration program to ensure system and equipment measurements are Safe, Accurate, Reliable, and Traceable to the National Institutes of Standards and Technology or other approved sources		
- Analyzes the health of PMEL's (e.g., Management Internal Control Test set, METWEB/Capability and Control, and Proficiency Testing/Measurement Assurance Program root causes, etc.) to prioritize the assignment of resources - Approves corrective action plans from Air Force Metrology and Calibration certification deficiencies		
- Monitors and controls the quality of PMEL products and processes using Quality Assurance data (e.g., Root Cause Analysis, Process Review Plan, Measurement Area Disciplines, and Process Reviews/Quality Reviews, etc.) to minimize problems or trends and report data within the applicable Maintenance Information System - Establishes a quality program to validate the technical proficiency, calibration process, and capability of the PMEL		
- Participates in Quality Assurance activities to assist with Quality Program execution - Conducts Quality Assurance functions to assess product quality and procedural compliance		

10.7.6. Another key component within the rubric is the supporting competencies section at the bottom left-hand corner. These are the top four supporting competencies that can help members excel and be successful in that particular sub-competency. Some of these supporting competencies are tied directly to the Airmen's Foundational Competencies, while others may be unique to the career field. Having these supporting competencies identified and linked to a career field's competency model can cultivate those underlying characteristics needed to succeed on the job. Leaders, supervisors, trainers, instructors, or mentors can now set members up for greater success by building these supporting competencies and placing their Airmen in situations where they can apply those strategies. All of these elements come together to ensure we can develop Airmen who are better prepared, present and future mission focused, and ready to succeed in any situation. Additionally, AFH 36-2643, *Air*

Force Mentoring Program, has information on how competencies can be used when an established mentoring strategy is put into effect to foster and develop Airmen.

10.7.7. The intent of moving towards a competency-based system is to sharpen our Airmen's tactical expertise, operational competence, strategic vision, and joint proficiency to lead and execute the full spectrum of USAF missions. This occurs not in a classroom but on the job by combining education, training, and experiences to provide Airmen with a better developmental pathway as they move along their careers. Airmen are still required to complete specific training courses, core tasks, and other training requirements in order to attain a 3-, 5-, and 7-skill levels. Competency development allows Airmen to move beyond the minimum career field requirements and begin addressing developmental gaps and strengthening their capabilities. The information included within the competency model will allow members within the Precision Measurement Equipment Laboratory community to manage their professional growth and development by identifying their strengths and weaknesses against clear and objective behaviors within the competency model.

10.7.8. Figure 7 below are the occupational competency rubrics for the 2P0X1 PMEL career field.

Figure 7. 2P0X1, The PMEL Occupational Competency Rubrics

Competency	Proficiency Levels	Observable Behaviors
Maintenance Activities	Expert Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency	- Develops new processes across Measurement Area Disciplines within the element to enhance the measurement capability - Collaborates with external agencies engineers to validate and develop alternate calibration methods
Sub-Competency	Advanced Consistency of Application: Sustained application of competency over time in complex situations	- Identifies deficiencies and takes corrective actions across multiple Measurement Area Disciplines within the Electronics Element - Determines deviations from procedural steps when substitute equipment is required - Maintains measurement standards traceability to Air Force Primary Standards Laboratory
Description	Intermediate Consistency of Application: Sustained application of competency over time in a variety of situations	- Maximizes measurements across multiple Measurement Area Disciplines within the Electronics Element to calibrate Test, Measurement and Diagnostic Equipment - Calculates tolerances of measurement results to determine the Test, Measurement and Diagnostic Equipment condition
Supporting Competencies	Basic Consistency of Application: Sustained application of competency over time	- Performs measurements utilizing metrology principles to calibrate Test, Measurement and Diagnostic Equipment - Utilizes measurement standards to apply established tolerances to maintain established test uncertainty ratios and determine Test, Measurement and Diagnostic Equipment condition - Applies general maintenance practices to produce safe, accurate, reliable, and traceable Test, Measurement and Diagnostic Equipment
Analytical Thinking Critical Thinking Precision Information Seeking		

Competency	Proficiency Levels	Observable Behaviors
Maintenance Activities	Expert Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency	- Develops new processes across Measurement Area Disciplines within the element to enhance the measurement capability - Collaborates with external agencies engineers to validate and develop alternate calibration methods
Sub-Competency	Advanced Consistency of Application: Sustained application of competency over time in complex situations	- Calibrates measurement standards that directly maintain laboratory traceability to Air Force Primary Standards Laboratory - Determines deviations from procedural steps when substitute equipment is required - Identifies deficiencies across multiple Measurement Area Disciplines within the Physical/Dimensional Element to calibrate Test, Measurement and Diagnostic Equipment
Description	Intermediate Consistency of Application: Sustained application of competency over time in a variety of situations	- Performs measurements across multiple Measurement Area Disciplines within the Physical/Dimensional Element to calibrate Test, Measurement and Diagnostic Equipment - Calculates tolerances of measurement results to determine the Test, Measurement and Diagnostic Equipment condition
Supporting Competencies	Basic Consistency of Application: Sustained application of competency over time	- Performs measurements utilizing metrology principles to calibrate Test, Measurement and Diagnostic Equipment - Utilizes measurement standards to apply tolerances to maintain test uncertainty ratios and determine Test, Measurement and Diagnostic Equipment condition - Applies general maintenance practices to produce safe, accurate, reliable, and traceable Test, Measurement and Diagnostic Equipment
Analytical Thinking Critical Thinking Precision Decision Making		

Competency	Proficiency Levels	Observable Behaviors
Maintenance Activities	Expert Depth of Knowledge: New practices/concepts and theories of all workplace elements; is a credible resource in this area	- Collaborates with external agencies to overcome troubleshooting and repair limitations - Develops a troubleshooting strategy to maximize effective resource management
Sub-Competency	Advanced Depth of Knowledge: New practices of all workplace elements	- Modifies Test, Measurement and Diagnostic Equipment to facilitate equipment modernization and sustainment strategies - Provides technical assistance for repairs and alignments - Validates troubleshooting techniques to resolve equipment deficiencies - Determines repair procedures and alignment techniques for Test, Measurement and Diagnostic Equipment not outlined in Technical Order or commercial data
Description	Intermediate Depth of Knowledge: Established practices of all workplace elements	- Applies troubleshooting techniques to resolve equipment deficiencies - Applies repair procedures and alignment techniques for Test, Measurement and Diagnostic Equipment
Supporting Competencies	Basic Depth of Knowledge: Established practice with some workplace elements	- Explains various troubleshooting techniques to resolve equipment deficiencies - Follows tech data to nominalize a performance specification - Applies general maintenance practices (e.g., Tools, PPE, ESD, etc.) to maintain equipment serviceability
Analytical Thinking Information Seeking Critical Thinking Flexibility		

Competency	Proficiency Levels	Observable Behaviors
Management System	<i>Expert</i> Reach of Influence: HAF/MAJCOM	- Establishes training requirements to ensure implementation of training programs for assigned DAF specialties - Monitors AFSC evaluation activities to ensure Airmen meet qualification requirements - Advocates for resources to support AFSC training and development
Sub-Competency	<i>Advanced</i>	- Forecasts training priorities to ensure an agile qualified workforce to meet mission requirements
Training	Reach of Influence: MAJCOM	- Secures training resources (e.g., coordinates Mobile Training Team, acquires equipment etc.) to enhance proficiency across the command
Description	<i>Intermediate</i>	- Develops training strategies to enhance proficiency in assigned measurement area disciplines
The process of providing a structural foundation of skills, methods, and characteristics to develop a qualified workforce.	Reach of Influence: Flight	- Maximizes external training resources (e.g., Mobile Training Teams, engineering assistance visit, Original Equipment Manufacturer training, etc.) - Evaluates the trainer's knowledge, skills, abilities, and other characteristics to meet job performance standards
Supporting Competencies	<i>Basic</i>	- Implements training strategies (e.g., CFETP, Technical Orders, AFIs, Training Plans etc.) to develop qualified Airmen
Communication Develops People Leadership Accountability	Reach of Influence: Element	- Identifies training constraints within the assigned element to ensure a qualified workforce

Competency	Proficiency Levels	Observable Behaviors
Management Systems	<i>Expert</i> Scope: Integration with AF-level/within industry: MAJCOM	- Monitors regional maintenance support capabilities to meet Command's mission requirements - Liaises with inter/intra-command functionals to address Test, Measurement and Diagnostic Equipment support concerns
Sub-Competency	<i>Advanced</i>	- Directs calibration network management activities to support Test, Measurement and Diagnostic Equipment lifecycle sustainment strategies
Logistics	Scope: Integration with organizational strategies: Flight	- Manages customer relations program to obtain feedback on Test, Measurement and Diagnostic Equipment matters - Implements maintenance priority system operations to support mission requirements
Description	<i>Intermediate</i>	- Monitors PMEL Automated Management System for data integrity to ensure alignment with mission priorities
Manage and coordinate all logistical activities related to the movement, storage, and maintenance of TDME, including inventory management, transportation, asset tracking, and customer relations.	Scope: Integration with concerned areas: Element	- Facilitates customer relations through two-way feedback on all Test, Measurement and Diagnostic Equipment matters - Completes calibration network management activities to support Test, Measurement and Diagnostic Equipment lifecycle sustainment strategies - Ensures safe handling and Test, Measurement and Diagnostic Equipment accountability to maintain resource stewardship
Supporting Competencies	<i>Basic</i>	- Initiates maintenance forms and documentation to ensure traceability to National Institute of Standards and Technology
Communication Organizational Awareness Resource Management Accountability	Scope: Specific Area: Individual	- Uses Digital Literacy through the utilization of PMEL Automated Management System or alternate Maintenance information system to process Test, Measurement and Diagnostic Equipment for maintenance

Competency	Proficiency Levels	Observable Behaviors
Management System	<i>Expert</i> Impact on...HAF-level practices/within industry	- Conducts Air Force Metrology and Calibration assessments to determine the effectiveness and the capability to perform Safe, Accurate, Reliable and Traceable measurements to the National Institute of Standards and Technology or other approved sources - Implements the Air Force Metrology and Calibration program to ensure system and equipment measurements are Safe, Accurate, Reliable, and Traceable to the National Institutes of Standards and Technology or other approved sources
Sub-Competency	<i>Advanced</i>	- Analyzes the health of Precision Measurement Equipment Laboratory's (e.g., Management Internal Control Test set, METWEB/Capability and Control, and Proficiency Testing/Measurement Assurance Program root causes, etc.) to prioritize the assignment of resources - Approves corrective action plans from Air Force Metrology and Calibration certification deficiencies
Quality Program	Impact on...MAJCOM Management decisions	
Description	<i>Intermediate</i>	- Monitors and controls the quality of PMEL products and processes using Quality Assurance data (e.g., Root Cause Analysis, Process Review Plan, Measurement Area Disciplines, and Process Reviews/Quality Reviews, etc.) to minimize problems or trends and report data within the applicable Maintenance Information System - Establishes a quality program to validate the technical proficiency, calibration process, and capability of the PMEL
Members systematically assess to sample PMEL product quality, procedural compliance, and effectiveness.	Impact on...Flight Specific workplace projects	
Supporting Competencies	<i>Basic</i>	- Participates in Quality Assurance activities to assist with Quality Program execution - Conducts Quality Assurance functions to assess product quality and procedural compliance
Analytical Thinking Communication Information Seeking Accountability	Impact on...Element Specific workplace tasks	

Competency	Proficiency Levels	Observable Behaviors
Management Systems	<i>Expert</i> Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency	- Programs for command resources (manpower, funding, equipment, etc.) to support mission objectives - Monitors performance metrics (e.g., METWEB/Capability and Capacity (CAP2), Management Reviews, etc.) for assigned PMELs - Coordinate and provide data between MAJCOM and Air Force Metrology and Calibration for Laboratory Evaluations
Sub-Competency	<i>Advanced</i>	- Establishes maintenance activities to produce safe, accurate, reliable, and traceable Test Measurement & Diagnostic Equipment - Develops local policies in accordance with governing directives (DAFMAN 21-113, Technical Order 00-20-14) - Advocates for resources to support mission objectives - Implements risk management to support mission and repair network management (DAFMAN 21-113, Technical Order 00-20-14)
Supervision	Consistency of Application: Sustained application of competency over time in complex situations	
Description	<i>Intermediate</i>	- Enforces maintenance discipline and policy compliance for responsible activities - Manages maintenance activities to produce safe, accurate, reliable, and traceable Test Measurement & Diagnostic Equipment - Monitors Key Performance Indicators to assess and improve element functions (DAFMAN 21-113, Technical Order 00-20-14)
Individuals or teams lead to ensure efficient and effective operations and adherence to regulations that achieve organizational goals.	Consistency of Application: Sustained application of competency over time in a variety of situations	
Supporting Competencies	<i>Basic</i>	- Utilizes resources in assigned duty position to enhance mission accomplishments - Manages mission support functions to ensure resource availability - Executes maintenance activities (e.g., Quality Assurance, Operational Risk Management, production, logistics) to produce safe, accurate, reliable, and traceable Test Measurement and Diagnostic Equipment
Communication Decision Making Leadership Accountability	Consistency of Application: Sustained application of competency over time	

10.8. Initial Skills/Apprentice (3) Level Requirements. Initial and Entry level training that is developed and taught by AETC.

10.8.1. Completion of the Basic PMEL Metrology Course

10.9. Journeyman (5) Level Upgrade Requirements.

10.9.1. Awarded 3-Level

10.9.2. Completion of the Intermediate PMEL Metrology CDC

10.9.3. Completion of all 5-Level core tasks

10.9.4. There is no time-in-training requirement

10.10. Craftsman (7) Level Upgrade Requirements.

10.10.1. Awarded 5-Level

10.10.2. Completion of Advanced PMEL Metrology CDC

10.10.3. Completion of all 7-Level core tasks

10.10.4. There is no time-in-training requirement

10.11. Superintendent (9) Level Upgrade Requirements.

10.11.1. Awarded 7-Level

10.11.2. Completion of Expert PMEL Metrology CDC (See Note, Para 18.7)

10.11.3. Completion of all 9-Level core tasks

10.11.4. There is no time-in-training requirement

10.12. Proficiency Training.

10.12.1. Intermediate PMEL Metrology Course is recommended when a 5-level reaches the rank of SrA with 2 years' time in grade.

10.12.2. Advanced PMEL Management Course is recommended when a 7-level reaches the rank of TSgt with 2 years' time in grade.

10.12.3. Career field functional exportable training accessible via AFCDA.

COMMUNITY COLLEGE OF THE AIR FORCE (CCAF)

11. Enrollment in the Community College of the Air Force occurs upon completion of Basic Military Training. Community College of the Air Force provides the opportunity to obtain an associate of applied sciences degree. In addition to its associate degree program, Community College of the Air Force offers a number of certifications that can be found in their Community College of the Air Force general catalog.

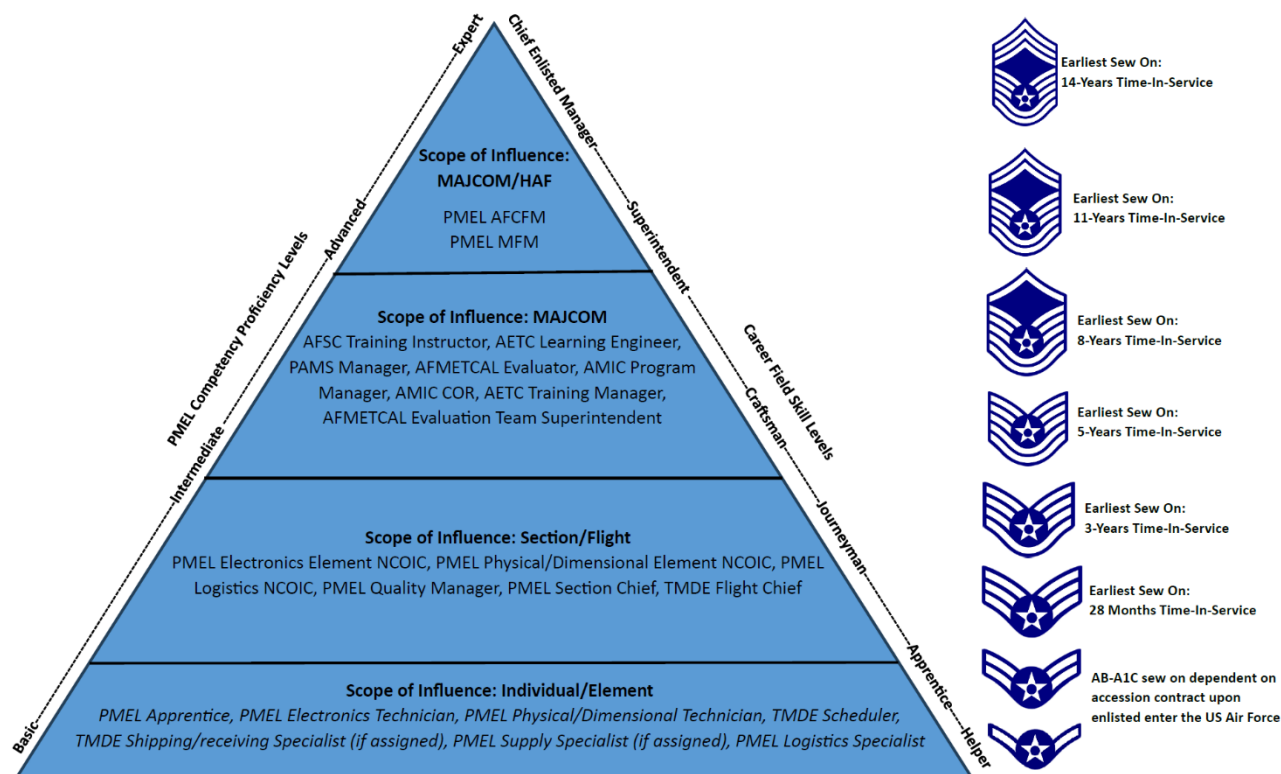
11.1. Degree Requirements. All service members are automatically entered into the CCAF program. Prior to completing an associate degree, the 5 level must be awarded and the following requirements must be met: (T-1) Note: Refer to the CCAF general catalog for a listing of all degree requirements as they are subject to change and some AFSCs may fall under additional degree programs.

Table 1. CCAF Degree Requirements

Subject Area	Semester Hours
Technical Education: A minimum of 12 semester hours of technical Core subjects and courses must be applied, and the remaining semester hours applied from technical core or technical elective subjects and courses.	24
Leadership, Management, and Military Studies: Professional military education and/or civilian management courses.	6
General Education: Applicable courses must meet the criteria for application of courses to the general education requirements and agree with the definitions of applicable general education subject and/or courses as provided in the CCAF general catalog.	15
Program Elective: Satisfied with applicable technical education; leadership, management, and military studies; or general education subjects and courses, including natural science courses meeting general education requirements application criteria. Six semester hours of CCAF degree applicable technical credit otherwise not applicable to this program may be applied. See the CCAF general catalog for details regarding the associates of applied sciences degree for this specialty.	15
Total	60

11.2. Additional off-duty education is a personal choice encouraged for all. Individuals desiring to become an AETC instructor should be actively pursuing an associate degree. A degreed faculty is necessary to maintain accreditation through the Southern Association of Colleges and schools Commission on Colleges.

11.3. Air Force Credentialing Opportunities On-Line (AF COOL) is a one-stop for Airmen to explore civilian credentialing opportunities that can enhance performance in their current Air Force specialty, as well as prepare for civilian employment. There are certain grade/specialty restrictions. For information, visit <https://afvec.us.af.mil/afvec/Public/COOL/>.

CAREER FIELD PATH**Table 2. Enlisted Career Path****12. Career Field Duty Titles**

12.1. PMEL Apprentice – 2P0X1 Airman who have graduated from the Initial Skills Course (Basic PMEL Metrology Course) and have been awarded the 2P031 AFSC. PMEL Apprentices will work with trainers to enhance their knowledge and skills to: Perform measurements utilizing metrology principles to calibrate TMDE. Utilize measurement standards to apply established tolerances to maintain established test uncertainty ratios and determine TMDE condition. Apply general maintenance practices to produce safe, accurate, reliable, and traceable TMDE.

12.2. PMEL Electronics Technician / PMEL Physical/Dimensional Technician – 2P051 Airman calibrating, repairing, or processing TMDE using 33K-series calibration procedures assigned to either the Electronic or Physical/Dimensional Elements.

12.3. TMDE Logistics Specialist – 2P0X1 Airman performing duties related to shipping, receiving, scheduling, supply and/or programs management. These duties may also be subdivided using the following duty titles: TMDE Scheduler, TMDE Shipping/Receiving Specialist, and TMDE Supply Specialist.

12.4. PMEL Quality Assurance Evaluator – 2P0X1 conducting quality assurance activities. QA Evaluators conduct inspections to ensure compliance with applicable regulations &

technical orders, accuracy of maintenance data collection in PAMS, evaluate TMDE condition and oversee compliance to PMEL cyber-security policy.

12.5. PMEL Electronics / PMEL Physical/Dimensional Supervisor - Use for 2P0X1 NCOs and above who rate on any AFSC 2P Airmen and/or provide training within the Electronics or Physical/Dimensional Element; may also be used for Senior Airman who have graduated Airman Leadership School and supervise Airmen.

12.6. PMEL Electronics Element NCOIC / PMEL Physical/Dimensional Element NCOIC – 2P051 or 2P071 NCOs responsible for daily production, training, and oversight of personnel predominately calibrating, repairing, and processing TMDE within the Electronics or Physical/Dimensional elements.

12.7. PMEL Logistics NCOIC – 2P051 or 2P071 NCOs responsible for oversight of personnel performing duties related to shipping, receiving, scheduling and/or programs management.

12.8. PMEL Quality Manager – 2P051 or 2P071 NCO or SNCOs responsible for management of the PMEL quality program activities, oversight of PMEL Quality Assurance Evaluators conducting inspections to ensure compliance with applicable regulations & technical orders, accuracy of maintenance data collection in PAMS, evaluating TMDE condition and overseeing compliance of PMEL cyber-security policy. Identifies risk within the PMEL to the TMDE Flight Chief.

12.9. PMEL Section Chief - The Section Chief is responsible for managing laboratory production demands to support the local mission and the PMEL Repair Network requirements. The Section Chief manages the PMEL training program and ensures maximum task coverage and technician proficiency.

12.10. TMDE Flight Chief - The senior on-site manager responsible for the overall PMEL, Management System, Quality Program, and Production Control functions. AFPSL management requirements are outlined in TO 00-20-14.

12.11. AMIC PMEL Contract Officer's Representative (COR) – 2P071 SNCOs/NCOs assist PMEL Air Force Enterprise Contracting Officers with development and management of contract requirements. They assist Contracting officers and Program Managers with contract invoicing and provide performance oversight ensuring PMEL Contractors meet the requirements of the contracts.

12.12. AMIC Air Force Enterprise PMEL Services Manager – 2P071 SNCOs responsible for monitoring PMEL Air Force Enterprise Contractor operations, personnel, and reimbursable TMDE support. They assist Program Managers and Contracting Officers in managing PMEL support program actions, contractor sponsorship, and alternate COR duties

12.13. AFSC Training Instructor – 2P071 NCOs and SNCOs responsible for delivering instruction system material at 2P0X1 formal training courses.

12.14. AETC Learning Engineer – 2P071 NCO responsible for the management and development of 2P0X1 Career Development Courses.

12.15. AETC Training Manager – 2P071 SNCO responsible for managing and directing the production, scheduling, and administration of all official AETC 2P0X1 training courses.

12.16. PAMS Manager – 2P071 SNCOs serving as 2P liaisons to system programmers and responsible for overseeing the PMEL Automated Management System (PAMS) system of record.

12.17. AFMETCAL Evaluator – 2P071 and 2P091 SNCOs responsible for conducting inspections in accordance with DAFI 90-302, *The Inspection System of the Department of the Air Force* and AFMAN 21-113.

12.18. AFMETCAL Evaluation Team Superintendent – 2P091 SNCO responsible for leading the AFMETCAL Evaluator team and serves as the assistant 2P0X1 AFCFM.

12.19. PMEL MFM – 2P071 and 2P091 SNCOs responsible for managing designated enlisted career fields and serving as the major command liaisons for their respective AFCFM.

12.20. PMEL AFCFM – 2P000 CMSgt responsible for oversight of the 2P0X1 career field, establishing career field entry requirements, managing trained personnel requirements, and developing and managing career-long training plan requirements and programs.

ENLISTED DEVELOPMENT TEAM (EDT)

13. The 2P0X1 EDT is the deliberate force development steering group for the 2P0X1 Air Force Specialty Code (AFSC) conducted in accordance with DAFI 36-2670, *Total Force Development*. The 2P0X1 EDT evaluates the training, education, and experience requirements for the most critical 2P0X1 AFSC Key Developmental Duty positions and provides assignment vector recommendations for the best qualified Senior Non-Commissioned Officers (SNCOs) into these key AFSC leadership positions at the unit, MAJCOM, and HHQ level. The 2P0X1 EDT also identifies other developmental opportunities for PMEL SNCOs to facilitate their deliberate development. These recommendations or vectors are the 2P0X1 EDT's collective recommendation for experience level, training and/or education opportunity, or position type that a member should be considered and seek out for professional growth. The recommendations do not take into consideration promotion factors. These vectors will consist of recommendations for identified positions (i.e. leadership and strategic positions) within the 2P0X1 AFSC construct for which a member should be considered in subsequent assignments but will not identify a specific location of assignment. For additional information or questions please refer to the 2P0X1 EDT Charter available through the assigned MAJCOM Functional Manager or 2P0X1 AFCFM.

SECTION C – SKILL LEVEL TRAINING REQUIREMENTS

PURPOSE

14. Skill level training requirements in this specialty are defined in terms of tasks and knowledge requirements. This section outlines the specialty qualification requirements for each skill level in broad, general terms and establishes the mandatory requirements for entry, award, and retention of each skill level. The specific task and knowledge training requirements are identified in the STS at Part 2, Section A and B of this CFETP.

SPECIALTY QUALIFICATION**15. Apprentice Level Training:**

15.1. Knowledge: The apprentice skill level is the base line learning level of PMEL Maintenance Activities and Management Systems competencies focusing on electrical & physical measurement principles, and logistics. Apprentice Airmen are expected to operate with direct supervisory & trainer oversight and assistance to obtain requisite qualification and proficiency to certify TMDE.

15.2. Education: For award of AFSC 2P031, completion of the Basic PMEL Metrology Course is mandatory.

15.3. Training: No specific OJT requirements needed for award of 3-skill level.

15.4. Experience: No special experience required for award of the 3-skill level.

15.5. Other: The following are mandatory as indicated:

15.5.1. For entry into this specialty:

15.5.1.1. Normal color vision as defined in AFI 48-123, *Medical Examinations and Standards*.

15.5.1.2. See the DAFECD, Attachment 4 for additional entry requirements.

15.5.2. For award and retention of the 2P0X1 AFSC:

15.5.2.1. Airmen must maintain local network access IAW AFI 17-130, *Cybersecurity Program Management* and AFMAN 17-1301, *Computer Security*.

15.6. Training Sources and Resources: Initial skills training encompasses metrology principles, TMDE theory and operation, measurement system components, component removal and installation, introduction to metrology concepts, use of technical publications, maintenance documentation, and support equipment familiarization and use.

15.7. Implementation: Upon graduation from Basic Military Training, Airmen are assigned to a training center for the completion of the Basic PMEL Metrology Course. Unless waived by the AFCFM, the Basic PMEL Metrology Course is a requirement upon entry into the career field.

16. Journeyman Level Training:

16.1. Knowledge: mandatory knowledge of PMEL Maintenance Activities and Management Systems competencies focusing on electrical & physical measurement principles, alignment/repair of TMDE, logistics, and supervision; analysis and interpretation of technical data, including block, schematic, wiring, and logic diagrams; troubleshooting techniques; metrology program, calibration traceability, metrology techniques, laboratory practices,

software, and computer operations principles; use of hand tools; procedures for training, supply, maintenance data collection, quality program, equipment scheduling, production, and material control; facility requirements; and support equipment management. Journeymen start to develop experience in functions such as troubleshooting, repairing, modifying, and aligning TMDE with limited assistance. Journeymen skill level is the steppingstone to training and development to calibrate more complex TMDE and laboratory reference standards.

16.2. Education: For award of AFSC 2P051, Airmen must possess the 2P031 AFSC and complete the Intermediate PMEL Metrology CDC.

16.3. Training: Completion of all OJT core tasks as outlined in Part 2.

16.4. Experience: No special experience required for award of the 5-skill level.

16.5. Other: Expected to maintain minimum requirements in the Apprentice skill level requirements listed above.

16.6. Training Sources and Resources: Career field standardized training products (e.g. Competency Assessment Workbooks) managed by the AFCFM and functional exportable training available through the AFCDA.

16.7. Implementation: Upon arrival at their first duty station, trainees are entered into UGT and OJT and must complete the requirements of both OJT and the 5-level CDC for award of the 5-skill level. Unless waived by the AFCFM, the Basic PMEL Metrology Course and the Intermediate PMEL Metrology CDC are requirements to be awarded the 5-skill level.

17. Craftsman Level Training:

17.1. Knowledge: knowledge of complex PMEL Maintenance Activities and Management Systems competencies focusing on electrical & physical measurement principles, alignment/repair of TMDE, training, logistics, supervision, and quality program across multiple measurement area disciplines; analysis and interpretation of technical data, including block, schematic, wiring, and logic diagrams; troubleshooting techniques; metrology program, calibration traceability, metrology techniques, laboratory practices, software, and computer operations principles; use of hand tools; procedures for training, supply, maintenance data collection, quality program, equipment scheduling, production, and material control; facility requirements; and support equipment management. Craftsmen serve as technical experts, trainers and supervisors at various levels within the organization. Additionally, Craftsman have an intermediate level of understanding of the PMEL Management System, Quality Program, AFMETCAL Program Office, and risk management applied per DAFMAN 21-113 and TO 00-20-14.

17.2. Education: For award of AFSC 2P071, Airmen must possess the 2P051 AFSC and complete the Advanced PMEL Metrology CDC.

17.3. Training: Completion of all OJT core tasks as outlined in Part 2.

17.4. Experience: No special experience required for award of the 7-skill level.

17.5. **Other:** Expected to maintain minimum requirements in the Journeyman skill level listed above.

17.6. **Training Sources and Resources:** Career field standardized training products (e.g. Competency Assessment Workbooks) managed by the AFCFM and functional exportable training available through the AFCDA.

17.7. **Implementation:** Once an individual is selected for promotion to SSgt, Airmen are entered into 7-level UGT and scheduled for the Advanced PMEL Metrology CDC. Trainees should utilize the CDC material to supplement knowledge needed for OJT.

18. Superintendent Level Training

18.1. **Knowledge:** Airmen awarded the 9-skill level have an expert level of understanding of the PMEL Management System, Quality Program, Air Force Metrology & Calibration Program Office, and risk management applied per DAFMAN 21-113 and TO 00-20-14. Airmen at this skill level have served in multiple roles within the career field at various levels of the career field. Airmen at this skill level have the requisite knowledge to establish training programs, support agreements, reimbursement procedures, and resource requirements.

18.2. **Education:** For award of AFSC 2P091, Airmen must possess the 2P071 AFSC and complete the Expert PMEL Metrology CDC.

18.3. **Training:** Completion of all OJT core tasks as outlined in Part 2.

18.4. **Experience:** No special experience required for award of the 9-skill level.

18.5. **Other:** Expected to maintain minimum requirements in the Craftsman skill level listed above.

18.6. **Training Sources and Resources:** Career field standardized training products (e.g. Competency Assessment Workbooks) managed by the AFCFM and functional exportable training available through the AFCDA.

18.7. **Implementation:** When an individual is selected for promotion to MSgt, Airmen are encouraged to begin qualification training on all 9-level core tasks. Once an individual is selected for promotion to SMSgt, Airmen are entered into 9-level UGT and scheduled for the Expert PMEL Metrology CDC. Trainees should utilize the CDC material to supplement knowledge needed for OJT. Note: Expert PMEL Metrology CDCs are in development. Once available, a myFSS training message (Article: 3423) will be sent out notifying the field and will include any additional information needed.

SECTION D – RESOURCE CONSTRAINTS

PURPOSE

19. This section identifies known resource constraints that preclude optimal and desired training from being developed or conducted, including information such as cost and manpower. Narrative explanations of each resource constraint and an impact statement describing what effect each constraint has on training are included. Also included in this section are actions required, OPR, and target completion dates. As a minimum, review and update resource constraints annually.

19.1. PMEL Apprentice Course. There are no constraints identified with this course. However, the course did experience temporary equipment & resource limitations during its launch and sustainment due to the lack of resource advocacy, effective POM/funding support, and TPM continuity within 2 AF/A3T. These shortfalls were mitigated by the career field through operational AFMETCAL program office channels.

19.2. PMEL Journeyman Distance Learning Course. There are no constraints identified with this course. This course has been deactivated and the Intermediate PMEL Metrology CDC has taken its place.

19.3. PMEL Craftsman Distance Learning Course. There are no constraints identified with this course. This course has been deactivated and the Advanced PMEL Metrology CDC has taken its place.

19.4. PMEL Craftsman In-Resident Course. This course has significant equipment constraints and cannot support STS training requirements. The constrained line items are identified with a “X” in the proficiency code column of the STS and will not be taught until the constraints are resolved. The course will launch as expected (first class Feb 2023) executing the unconstrained training line items only. A STRT is slated for 2Q FY23 to resolve the issue and determine a path forward. Per the FY23 STRT, the launch of the 7-lvl In-residence course has been suspended and the training efforts will be executed through a newly developed Intermediate PMEL Metrology Course.

19.5. Advanced PMEL Management Course. There are no constraints identified.

SECTION E – TRANSITIONAL TRAINING GUIDE

20. There are no transitional training requirements. This area is reserved.

PART II

SECTION A - SPECIALTY TRAINING STANDARD

21. Implementation. The STS – (attachment) will be used for initial skills training provided by AETC for classes beginning June 2025.

21.1. Job Qualification Standard (JQS). This STS becomes a job qualification standard (JQS) for OJT when placed in AF Form 623, *On-The-Job Training Record* or electronic equivalent. For OJT, the competencies in ‘Column A’ are trained and qualified to the go/no-go level. ‘Go’ means the individual can perform the competency without assistance and meets local requirements for accuracy, timeliness, and correct procedures.

21.2. Air Force Job Qualification Standard (AFJQS). The AFJQS identifies unique duty positions, weapons systems, or equipment that may not be adequately covered in the competencies listed in the STS. The AFJQS supplements the CFETP, Part 2, by outlining specific skill and task requirements. The AFJQS has been limited to the absolute minimum number of items to reduce duplicate documentation.

21.3. Documentation. Document and certify completion of training IAW DAFI 36-2670. Use of Part 2 and the attachment of this CFETP are mandatory in individual training records. As a minimum, complete the following columns in Part 2 of the CFETP: date training started, date training completed, trainee initials, and trainer initials. It is the work center supervisor’s responsibility to identify work center requirements and build a Master Training Plan (MTP) to develop assigned trainees to the requirements.

21.4. Members currently in Upgrade Training. Members who are currently in UGT and have been in UGT for greater than 12 months from the publication date of this CFETP will continue to complete UGT under the former CFETP’s requirements. Once upgraded, their training records will be transcribed per the new CFETP. Members who have been in UGT for less than 12 months from the publication date of this CFETP will convert over to the new CFETP requirements.

21.5. Transcribing. The previous CFETP and AFJQS are replaced by this CFETP; therefore, transcribing of training records to this CFETP STS is mandatory. Use this CFETP STS (or automated STS) to identify and certify all past and current qualifications. Transcribing documentation to a new CFETP is an administrative function, not a re-evaluation of training. Therefore, supervisor and trainer are considered synonymous for the purpose of documentation. Transcribe within 120 calendar days of CFETP revision date or from date revision is posted to automated training records system.

22. STS Description. (Attachment)

22.1. Column A lists (Competencies, Required Behaviors, Knowledge, and Technical References (TR)) necessary for Airmen to perform duties in the 3, 5, 7, and 9 skill levels.

22.2 Column B (Behavior Match) A number following the competency description indicates which of the required behaviors listed in the competency heading applies.

22.3 Column C (Core/Cert ^) A number in this column indicates a competency that is required for that particular skill level. AFSC 2P0X1 does not currently have any competencies which require 3rd party certification (which would be indicated with a ^ symbol).

22.4 Column D (Deployment */SEI +/CBRN ~) Competencies identified with an (*) are foundational competencies 2P0 technicians should be qualified on (per their skill level) prior to deployment. Competencies identified with a (+) are reserved for award of a specific SEI. Currently no competencies are identified with a (+). Competencies identified with an (~) require annual CBRN (Training Task Qualification) training in the work center. Per DAFI10-2503, para 6.5.1.1., CBRN Defense TQT is defined as a hands-on event in MOPP 4 gear performing regular duties. At a minimum, individuals will be evaluated on their ability to accomplish AFSC-specific competencies (according to section of assignment (ie. Electronics, Logistics, etc.) while wearing CBRN defense IPE in MOPP 4. NOTE: All assigned 2P0 personnel are eligible to perform TFCU deployment packout tasks while wearing CBRN defense IPE in MOPP 4, if workcenter is assigned a TFCU. Document CBRN training and recurring evaluations through a journal entry in MyTraining, TBA, or local equivalent.

22.5. Column E (Wartime) Competencies with a (W) in this column indicate items that must be taught in the 3-level technical training school when courses are accelerated in a wartime environment.

22.6. Columns K-P Shows formal training and correspondence course requirements. These are the proficiencies to be demonstrated on the job by the graduate as result of training on the competency/knowledge and the career knowledge provided by the in-residence or correspondence course (see Behavioral Key Code tab).

23. SKT Development. STSs are used as guides for the development of promotion tests used in the WAPS. Specialty Knowledge Tests (SKTs) are developed at the AETC A2/9 SAS-OA (Airman Advancement Division) by NCOs or SNCOs with extensive practical experience in the career field. The tests sample knowledge of STS subject matter areas judged by test development team members to be most appropriate for promotion to higher grades. Questions are based on study references listed in the Enlisted Promotions References and Requirements Catalog (EPRRC).

SECTION B - COURSE OBJECTIVE LIST - A detailed listing of initial, intermediate, or advanced skills course objectives is available upon request; contact the OPR.

SECTION C - SUPPORT MATERIAL - There are currently no support material requirements. This area is reserved for future utilization as necessary.

SECTION D - TRAINING COURSE INDEX - This section identifies training courses available for the 2P0X1 specialty. Refer to the Air Force Education and Training Course Announcements (ETCA) for information on AETC formal courses listed here: [ETCA Course Search: Course ID: 2P0](#)

SECTION E – MAJCOM-UNIQUE REQUIREMENTS - There are currently no unique MAJCOM requirements. This area is reserved.

ABBREVIATIONS/TERMS EXPLAINED

Advanced Training: Formal course, which provides individuals who are qualified in their Air Force Specialty with additional skills and/or knowledge to enhance their expertise in the career field. Training is for selected career airmen at the advanced level of an AFS.

Air Force Job Qualification Standard: A comprehensive task list that describes a particular job type or duty position. Supervisors use the AFJQS to document task qualifications. The AFJQS tasks are common to all persons serving in the described duty position.

Career Field Education and Training Plan: A CFETP is a comprehensive, multipurpose document covering the entire spectrum of education and training for a career field. It outlines a logical growth plan that includes training resources and is designed to make career field training identifiable, eliminate duplication, and ensure this training is budget defensible.

Career Development Course: CDCs are published to provide the information necessary to satisfy the career knowledge component of OJT training. These courses are developed from references identified in the CFETP correlating with mandatory knowledge items listed in enlisted or officer classification directory. CDCs contain information on basic principles, techniques, and procedures common to an AFSC. They do not contain information on specific equipment or tasks unless best illustrating a procedure or technique having utility to the entire AFSC. Members will adhere to all procedures for CDC administration that are outlined in the UTM CDC administration PSD guide.

Certification: A formal indication of an individual's ability to perform a task to required standards.

Certification Official: A person authorized by appropriate commander to determine an individual's ability to perform a task to required standards.

Continuation Training: Additional training that exceeds minimum upgrade requirements and has an emphasis on present or future duty assignments.

Core Task: Tasks that the Air Force Career Field Manager identifies as minimum qualification requirements within an Air Force Specialty.

Exportable Training: Additional training via computer assisted, paper text, interactive video or other necessary means to supplement training.

Go/No Go Level: In On-the-Job Training, the stage at which individual has gained enough skill, knowledge, and experience to either be qualified to perform identified task without assistance or cannot perform task without assistance.

Initial Skills Training: A formal school course that results in the award of a 3-skill level Air Force Specialty Code.

Mobile Training Team: Instructors, trainers, training aids, and operational equipment that formal schools send to bases or operating locations used to perform formal training.

Occupational Analysis Report: A detailed report showing the results of an occupational survey of tasks performed within a particular Air Force Specialty.

On-the-Job Training: Hands-on, over-the-shoulder training at the duty location used to certify personnel for both skill level upgrade and duty position qualification.

Proficiency Training: Additional training provided via in-residence courses, exportable advanced training courses, or On-the-Job Training to personnel to increase their skills and knowledge beyond the minimum required for upgrade.

Qualification Training: Actual hands-on task performance training designed to qualify an Airman in a specific duty position. This training program occurs both during and after the upgrade training process. It is designed to provide the performance skill/knowledge training required to do the job.

Qualification Training Package: An instructional course designed for use at the unit to qualify or aid qualification in a duty position or program or on a piece of equipment. It may be printed, computer based, or in other audiovisual media.

Resource Constraints: Resource deficiencies such as money, facilities, time, manpower, or equipment that preclude desired training from being accomplished.

Specialty Training Standard: An Air Force document that is published as an attachment to the appropriate CFETP that describes an Air Force Specialty in terms of tasks and knowledge an Airman may be expected to perform or to know on the job. It serves as a contract between AETC and the functional user to show which of the overall training requirements for an Air Force Specialty Code are taught in formal schools, in job knowledge courses, and exportable courses.

Specialty Training Requirements Team: Primary purpose of the STRT is for the Air Force Career Field Manager and MAJCOM Functional Managers to determine and present training requirements to AETC. Attendees include the AFCFM, AETC Training Pipeline Manager, AETC Training Managers, MFMs, Subject Matter Experts, and AETC Occupational Analysis Division. The AFCFM chairs the STRT. The AETC TPM and TMs attend as advisors.

Supplemental Training: Formal, standardized training within an Air Force Specialty that is in addition to required initial skills training and skill level upgrade training. It may support new and/or newly assigned equipment, methods, and/or technology.

Task Certifier: See Certification Official

Training Detachment: An AETC detachment that provides maintenance oriented technical training at an operational location. Training can address specific systems, including aerospace ground equipment, or cover new equipment techniques and procedures. A TD qualifies personnel to maintain proficiency, increase skill and knowledge, acquaint personnel with specific systems, and keep personnel aware of changing concepts and requirements.

Training Setting: The type of forum in which training is provided (formal in-residence school, On-the-Job Training, field training, Mobile Training Team, self-study, etc.).

Upgrade Training: A mixture of mandatory courses, task qualification, Qualification Training Packages, and job knowledge courses required for award of a 3-, 5-, 7-, or 9-skill level.

Utilization and Training Workshop: A forum convened and chaired on a recurring basis by the AF Career Field Manager, designed to review the appropriate Career Field Education and Training Plan and its attachments. The purpose is to ensure currency, accuracy, and completeness of content, to include specific formal career ladder training requirements. Workshops are co-chaired by AETC Training Pipeline Manager and include MAJCOM Functional Managers, AETC training personnel, and Subject Matter Experts.

ACRONYMS

AFCDA – Air Force Career Development Academy
AFCFM – Air Force Career Field Manager
AF-COOL – Air Force Credentialing Opportunities On-Line
AFJQS – Air Force Job Qualification Standard
AFS – Air Force Specialty
AFSC – Air Force Specialty Code
AFTC – Air Force Training Course
AFVEC – Air Force Virtual Education Center
ALS – Airman Leadership School
AMIC – Acquisition Management Integration Center
AU – Air University
BMT – Basic Military Training
CCAF – Community College of the Air Force
CDC – Career Development Course
CFETP – Career Field Education and Training Plan
DAFECD – Department of the Air Force Enlisted Classification Directory
DL – Distributed Learning
DLC – Distance Learning Course
EPME – Enlisted Professional Military Education
ISD – Instructional System Development
ITU – Instructional Technology Unit
JQS – Job Qualification Standard
MFM – Major Command Functional Manager
MTL – Master Task Listing
MTP – Master Training Plan
MTT – Mobile Training Team
NCOA – Noncommissioned Officer Academy
NCOIC – Noncommissioned Officer in Charge
OAR – Occupational Analysis Report
OJT – On-the-Job Training
PMC – Professional Manager Certification
QA – Quality Assurance
SME – Subject Matter Expert
SNCOA – Senior Noncommissioned Officer Academy
STRT – Specialty Training Requirements Team
STS – Specialty Training Standard
TD – Training Detachment
TIS – Time in Service
TM – Training Manager
TO – Technical Order
TPM – Training Pipeline Manager

TR – Training Resource
UGT – Upgrade Training
UTC – Unit Type Code
UTM – Unit Training Manager